113223

103963

5. L em série com a linha

L= 8.438 nH

d=0.035 m

Z_o

Z_L

$Z_L = 71.1 + 51.1j$ (Ω)

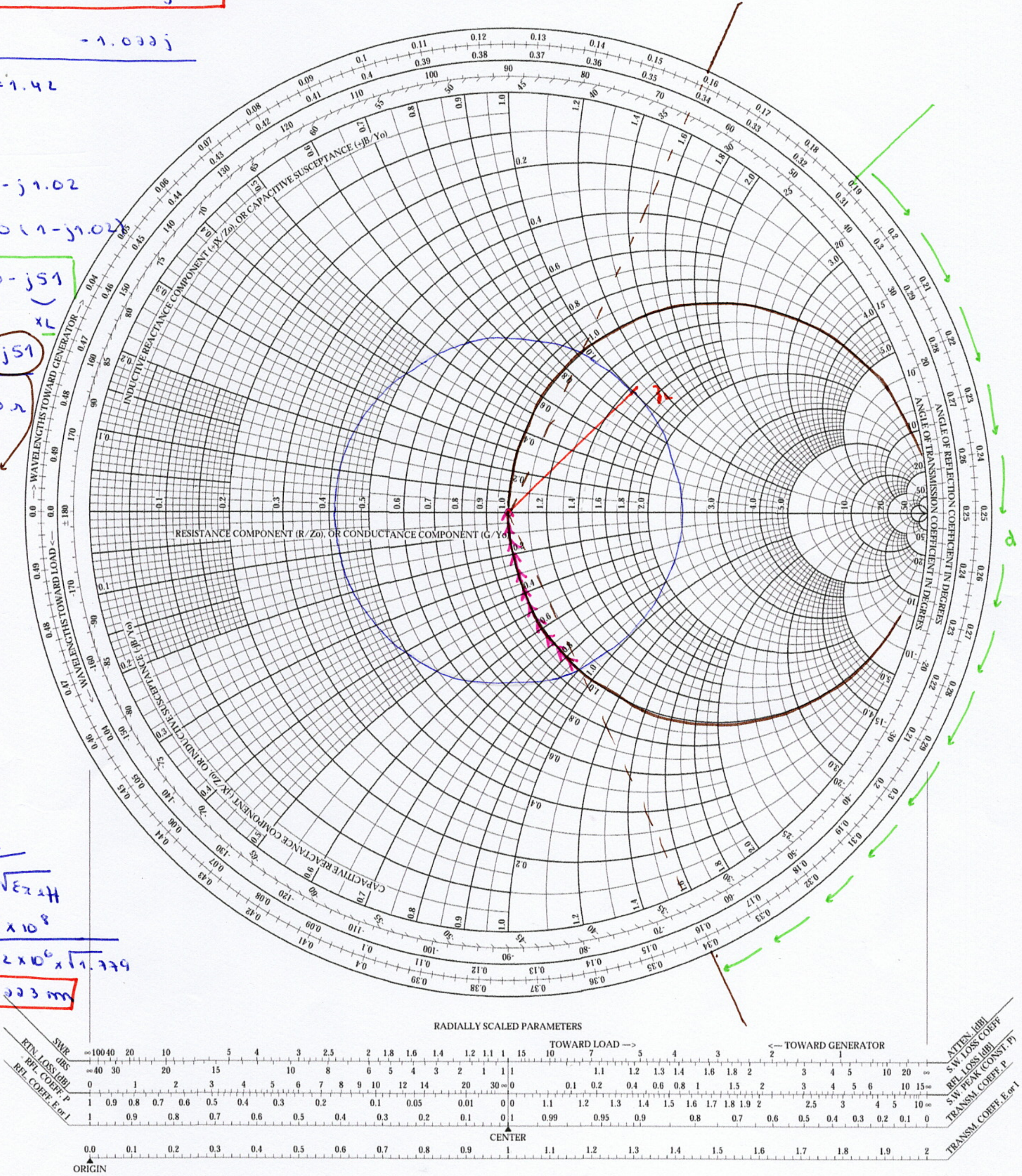
$\boxed{z_L = 1.42 + 1.022j \ (\Omega)}$

$-1.022j$

$z = 1.42$

The Complete Smith Chart

Black Magic Design



$z = 1 - j1.02$

$Z = 50(1 - j1.02)$

$\boxed{Z = \underbrace{50}_{R} - \underbrace{j51}_{X_L}}$

$+j51$

bobine

$Z = 50\ \Omega$

$\lambda = \dfrac{c}{f\sqrt{\varepsilon_r \mu_r}}$

$= \dfrac{3\times10^8}{962\times10^6 \times \sqrt{1.779}}$

$\boxed{\lambda = 0.233\ m}$

$d = 0.341\lambda - 0.189\lambda$

$d = 0.152\lambda$

$\boxed{d = 0.035\ m}$

$L = \dfrac{X_L}{j\omega} = \dfrac{51}{2\pi \times 962\times10^6}$

$L = 8.438\times10^{-9}$

$\Leftrightarrow \boxed{L = 8.438\ nH}$